Derivatives So Far

MTH 207 — through §3.6

There is **no formula sheet** for any of this. Shaded rows are the ones to **memorize**. **DERIVE** means you may be asked to *derive* the rule rather than quote it — know where it comes from. Rows in *gray italics* you never have to learn.

Powers, exponentials, logs

$f(x)$	$f'(x)$	
c	0	**MEMORIZE**
x^n	$n\,x^{n-1}$	**MEMORIZE**
e^x	e^x	**MEMORIZE**
a^x	$a^x \ln a$	**MEMORIZE** + **DERIVE**
$\ln x$	$\dfrac{1}{x}$	**MEMORIZE**
$\log_a x$	$\dfrac{1}{x \ln a}$	**MEMORIZE**

Rebuild a^x and $\log_a x$ by rewriting: $a^x = e^{x \ln a}$, $\log_a x = \dfrac{\ln x}{\ln a}$.

Trigonometric

$f(x)$	$f'(x)$	
$\sin x$	$\cos x$	**MEMORIZE**
$\cos x$	$-\sin x$	**MEMORIZE**
$\tan x$	$\sec^2 x$	**MEMORIZE** + **DERIVE**
$\cot x$	$-\csc^2 x$	**MEMORIZE** + **DERIVE**
$\sec x$	$\sec x \tan x$	**MEMORIZE** + **DERIVE**
$\csc x$	$-\csc x \cot x$	**MEMORIZE** + **DERIVE**

Derive the last four by writing each in sin and cos and using the Quotient Rule. The "co" functions are the ones whose derivatives carry a minus sign.

Inverse trigonometric — the memorization load is **two**

$f(x)$	$f'(x)$	
$\sin^{-1} x = \arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$	**MEMORIZE** + **DERIVE** (draw the triangle for $\sin y = x$)
$\tan^{-1} x = \arctan x$	$\dfrac{1}{1+x^2}$	**MEMORIZE** + **DERIVE** (draw the triangle for $\tan y = x$)
$\cos^{-1} x = \arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$	flip the sign of the one above (arcsin): $\arcsin x + \arccos x = \frac{\pi}{2}$
$\cot^{-1} x = \operatorname{arccot} x$	$-\dfrac{1}{1+x^2}$	flip the sign of the one above (arctan): $\arctan x + \operatorname{arccot} x = \frac{\pi}{2}$
$\sec^{-1} x$	*$\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$*	*reference only — never asked*
$\csc^{-1} x$	*$-\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$*	*reference only — never asked*

The rules that combine them — memorize all five

Constant multiple	$\dfrac{d}{dx}\big[c\,f(x)\big] = c\,f'(x)$
Sum and difference	$\dfrac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x)$
Product	$\dfrac{d}{dx}\big[f(x)\,g(x)\big] = f'(x)\,g(x) + f(x)\,g'(x)$
Quotient	$\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{f'(x)\,g(x) - f(x)\,g'(x)}{\big[g(x)\big]^2}$
Chain	$\dfrac{d}{dx}\Big[f\big(g(x)\big)\Big] = f'\big(g(x)\big) \cdot g'(x)$

Logarithm properties — memorize

$$\ln(AB) = \ln A + \ln B$$

$$\ln \frac{A}{B} = \ln A - \ln B$$

$$\ln\left(A^n\right) = n \ln A$$

$$\ln 1 = 0, \qquad \ln e = 1$$

$$e^{\ln x} = x, \qquad \ln\left(e^x\right) = x$$

$$\log_a x = \frac{\ln x}{\ln a}$$

For $A, B > 0$. **Not a law:** $\ln(A + B)$ — a log of a *sum* does not expand.

Two methods, not formulas. Implicit (§3.5): differentiate both sides; every y carries a $\frac{dy}{dx}$. **Logarithmic** (§3.6): take ln of both sides, *expand with the log laws first*, then differentiate — the way to handle x^x.